Counting maximal-subgroup pairs from fixed points

This note supplies reductions for the remaining claims of Leemans and Rodrigues, *On primitive geometries of rank two*, Art Discrete Appl. Math. 2 (2019), P1.08. Numerical applications and their exact data are supplied separately. The source's equivalence is the action of $\mathrm{Aut}(G)$ on unordered pairs of distinct maximal subgroups of G. A pair determines a firm, connected, flag-transitive primitive coset geometry. Conversely, the source's geometries have precisely this form. These facts and the Rudvalis lower bound are already proved in the accepted principal correction.

1 The conjugation action, including outer automorphisms

For a finite group A, choose representatives $H_1, \dots, H_r$ of all conjugacy classes of subgroups. Write

$$m_{ij} = |(A/H_i)^{H_j}|.$$

Thus the rows of the table of marks give fixed points on transitive A-sets. In particular, $m_{ij} > 0$ if and only if H_j is contained in an A-conjugate of H_i. Also

$$m_{i1} = [A : H_i], \qquad m_{ii} = [N_A(H_i) : H_i],$$

where $H_1 = 1$. The first equality follows by taking fixed points of the trivial group. For the second, a coset fixed by H_i corresponds to a conjugate of H_i containing H_i; equal orders force equality, and the set of such cosets is $N_A(H_i)/H_i$.

Lemma 1. *Let G be a finite nonabelian simple group and let $G \le A \le \mathrm{Aut}(G)$, where G is identified with its inner automorphism group. Suppose $[A : G] \le 2$. If M is maximal in G, then $N_A(M)$ is determined, up to A-conjugacy, by the orders and containment relation in the table of marks of A.*

Proof. The group M is nontrivial: a nonabelian simple group cannot have the trivial group as a maximal subgroup. Maximality gives $N_G(M) = M$ or G. The latter would make M a nontrivial proper normal subgroup, so $N_G(M) = M$. The map $N_A(M) \to A/G$ has kernel M. Consequently $[N_A(M) : M]$ is one or two. This index is the diagonal mark at M. If it is one, the normalizer is M.

Suppose it is two. Any subgroup K of order $2|M|$ containing M normalizes M, because a subgroup of index two is normal. Hence $K \le N_A(M)$, and equal orders give $K = N_A(M)$. At least one such subgroup exists, namely the normalizer itself. Thus precisely one conjugacy class of subgroups of order $2|M|$ can contain a conjugate of M. Both its existence and its class are read from the table. □

If $[A : G] = 2$, the class of G is the unique subgroup class of order $|A|/2$. Indeed, any subgroup K of that index is normal in A. The intersection $G \cap K$ is normal in G and has index at most two there. Simplicity and $|G| > 2$ imply $G \le K$, and then $G = K$. The maximal subgroups of G, up to A-conjugacy, are exactly the maximal proper classes below this class in the containment relation. Because G is normal, containment in a conjugate of G is simply containment in G. If $A = G$, use the final class instead.

Theorem 2. *Under the hypotheses of Lemma 1, let $\mathcal{M}$ contain one representative of every A-conjugacy class of maximal subgroups of G. For a conjugacy class representative $a \in A$, put*

$$\psi(a) = \sum_{M \in \mathcal{M}} m_{N_A(M),\langle a \rangle},$$

where the subscripts denote the corresponding subgroup classes. Then the number of A-orbits on unordered pairs of distinct maximal subgroups of G is

$$\frac{1}{2} \sum_{[a]_A} \frac{\psi(a)^2 - 2\psi(a) + \psi(a^2)}{|C_A(a)|}. \tag{1}$$

Thus a complete table of marks, conjugacy-class centralizers, a square map, and the fusion taking an element class to its cyclic subgroup class determine this number by exact integer arithmetic.

Proof. The conjugation orbit of M is the transitive A-set $A/N_A(M)$. Its points fixed by a are its points fixed by $\langle a \rangle$, hence are counted by the displayed mark. The disjoint union of these orbits is the set of all maximal subgroups of G. Therefore $\psi(a)$ is its fixed-point count.

An unordered pair fixed by a is either a pair of fixed points or a two-cycle. There are $\binom{\psi(a)}{2}$ of the first kind and $(\psi(a^2) - \psi(a))/2$ of the second. Burnside's orbit-counting lemma, grouped by element conjugacy classes, gives (1), since the class of a has size $|A|/|C_A(a)|$. □

The formula uses the full action of A. Replacing it by the action of G when outer automorphisms are present does not compute the source's isomorphism classes.

When $A = G$, a smaller character-table certificate suffices. Since $N_G(M) = M$, the required action is the union of the coset actions G/M. If the classes of M fuse to those of G, then

$$|(G/M)^g| = |C_G(g)| \sum_{[h]_M \mapsto [g]_G} \frac{1}{|C_M(h)|}. \tag{2}$$

To prove (2), count the elements $x \in G$ with $x^{-1}gx \in M$. Each suitable element of M has $|C_G(g)|$ preimages. Dividing by $|M|$ counts cosets; expanding the suitable elements into M-classes gives the stated expression.

2 Suborbit lengths from products of marks

Theorem 3. *Let $H, K \leq G$ and let m_L denote the mark row of G/L. There are unique nonnegative integers c_L, indexed by the conjugacy classes of subgroups of G, such that*

$$m_H \, m_K = \sum_{[L]_G} c_L m_L, \tag{3}$$

where multiplication on the left is coordinatewise. The multiset of orbit lengths of H on G/K consists of c_L copies of $|H|/|L|$ for each L with $c_L > 0$. In particular, $|L|$ divides $|H|$ for every such coefficient.

Proof. Fixed points commute with Cartesian products. Thus the left side of (3) is the fixed-point vector of the diagonal action of G on $G/H \times G/K$. Decompose this finite G-set into its transitive orbits. Each orbit is isomorphic to G/L for the stabilizer of one of its points; collecting conjugate stabilizers gives (3) with nonnegative integral coefficients.

Order the subgroup representatives by increasing order, with an arbitrary order among equal orders. Containment makes the table of marks triangular, with nonzero diagonal. Its

rows are linearly independent over $\mathbb{Q}$, proving uniqueness and giving an exact triangular solution for (3).

Every diagonal orbit meets $\{H\} \times G/K$. Its intersection with this fiber is one H-orbit: if a group element sends a point of the fiber to another, it fixes the first coordinate and belongs to H. The stabilizer of (H, gK) is $H \cap gKg^{-1}$. Hence the corresponding H-orbit has length $|H|/|H \cap gKg^{-1}|$. Its stabilizer is conjugate to the L attached to the diagonal orbit, proving the formula and divisibility assertion. □

This theorem recovers the full multiset of suborbit lengths, including singletons. For two conjugacy classes of maximal subgroups, singleton orbits occur precisely when the two actual subgroups coincide. The suborbit multiset alone does not identify which orbits are interchanged by duality or by outer automorphisms; Theorem 2 handles that separate equivalence. Nor does it recover diameters, girths or design parameters of the individual incidence graphs.

3 Attribution and remaining certificate obligations

Tables of marks, their injective fixed-point map, the product (3), induction of trivial characters, and Burnside's lemma are classical. Our proposed contribution is their application to the source's complete list of counting and suborbit claims, with an explicit normalizer step for the outer automorphisms. The GAP Table of Marks Library supplies a complete subgroup table as an attributed premise; internal consistency checks do not prove that the table is complete or belongs to the named group. The Character Table Library supplies the named character tables, complete maximal-subgroup list and compatible fusions when (2) is used. The installed documentation explicitly states this meaning of `Maxes`. All library editions and selected files must be bound before a numerical application is accepted.

The accepted M_{22} census in the existing L073 recovery already gives 81 pair classes under the required full automorphism equivalence. It may be cited without redoing its enumeration. The companion application verifies five character totals by independent exact arithmetic; it requires its own distinct review. The source's 37 M_{11} incidence graphs and their design parameters still require actual representatives and finite checks. Reported historical timings are observations of the original implementation, not mathematical claims recovered by the fixed-point formulas.